



SANDEEP MERUGUMALA

Roll No.:CB.SC.U4CSE23664

B.Tech - Computer Science and Engineering

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[Github](#) | [Website](#)

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EDUCATION

Degree/Certificate	Institute/Board	Score	Year
B.Tech	Amrita Vishwa Vidyapeetham, Coimbatore	7.34/10.0 (Current)	2023-Present
Intermediate	State Board	96.9%	2023
Secondary	State Board	10.0/10.0	2021

PROJECTS

•ShopsBuzz – Online Shopping Website

Node.js, Express.js, MongoDB, Razorpay API

[Link](#)

- Developed the backend architecture using Node.js to manage core application logic and enable smooth interaction with the frontend.
- Integrated Razorpay Payment Gateway to provide secure and reliable payment processing.
- Implemented APIs for authentication, product management, and order workflows.

•Maze Pathfinding System using A* Search and Fibonacci Heap

Python

[Github](#)

- Engineered a Python-based maze generator and A* pathfinding engine, achieving 100% path accuracy across dynamically generated grids up to 1,200+ cells.
- Developed a custom Fibonacci Heap for priority-queue optimization, resulting in 30–40% faster frontier operations compared to standard binary heap structures.

•Smart Irrigation System (STM32)

STM32F103C8T6, Embedded C, Soil Moisture Sensor

[Github](#)

- Designed an automated irrigation system that monitors soil moisture levels in real-time.
- Implemented threshold-based pump activation logic to reduce unnecessary water usage.

SKILLS

- **Area Of Interest:** Frontend Development, Web Development, DBMS, Mobile App Development, Embedded Systems
- **Languages and Frameworks:** C, Java, Python, MySQL, HTML5, CSS3, JavaScript, NodeJS, ExpressJS, ReactJS, Flutter, Dart, Firebase
- **Operating Systems Tools:** Windows, Linux, Git/GitHub, Webots, Keil uVision (STM32), VS Code, Wireshark
- **Non Technical:** Organization, Financial Analysis, Time Management, Problem Solving

LICENSES & CERTIFICATION

- AWS Academy Graduate - Cloud Foundations, [View Certificate](#)
- Data Science for Engineers, [View Certificate](#)

EXTRACURRICULARS

•Student Volunteer – National Service Scheme (NSS)

- Contributed to campus-level community service including event coordination, environmental awareness programs, and volunteer support activities.